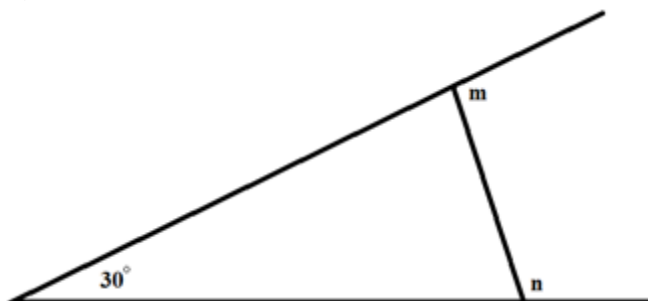


1.) $\frac{\log_4 64}{4^2} = ?$

2.) The value of $m+n=?$



CASIO®

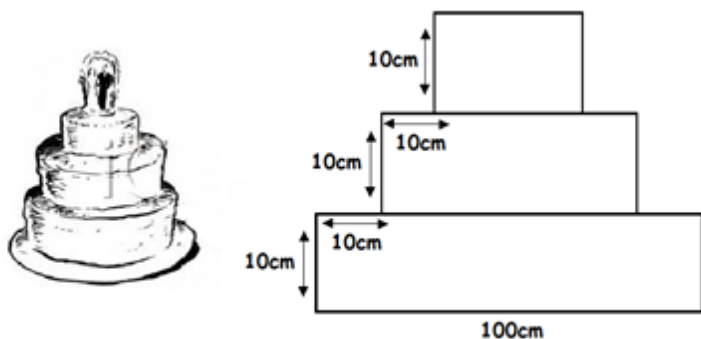
St David's Marist Inanda

Tour de Maths 2014 Senior Paper

3.) The length and width of a rectangle are both doubled. In percentage form, what will there be to BOTH the area and the perimeter.

4.) $2^x + 3^y = 41$ If both x and y are natural numbers.
 $x + y = ?$

5.) A hollow, 3 tiered circular cake is filled with icing at a constant rate of 1 ℓ /hr. The bottom tier has a height of 10cm and a diameter of 100cm. Each tier has a diameter which is 20cm smaller than the one before and same height. After how many minutes will the cake be filled with icing?



6.) The first 10 prime numbers add up to?

7.) The remainder of 3^{1998} when divided by 5 is?

8.) If a bat costs R10 more than a ball and together they cost R110, how much does the bat cost?

9.) x is an integer such that,

$$9 < x^2 < 99$$

What is the maximum x value – minimum x value=?

10.) The maximum value of $\sin \theta + \cos \theta = ?$

11.) The sum of two numbers is 10. The difference of their squares is 40. The sum of their squares is?

12.) How many different ways are there of making the word 'CASIO' making strings of letters from top to bottom.

C
A A
S S S
I I
O

13.) How many of the following numbers are greater than 10?

$$3\sqrt{11} \quad 4\sqrt{7} \quad 5\sqrt{5} \quad 6\sqrt{3} \quad 7\sqrt{2}$$

14.) How many 4 digit numbers are there such that the thousands digit is equal to the sum of the other 3 digits?

15.) The sequence 1234567891234567891234567... is continued until the sum of all the digits used is 460. The last digit in the sequence is?

16.) The value of

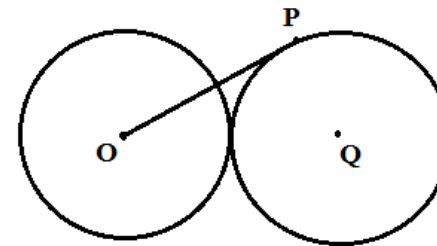
$$\frac{1 \times 2 \times 4 + 2 \times 4 \times 8 + 3 \times 6 \times 12 + \dots + 10 \times 20 \times 40}{1 \times 3 \times 9 + 2 \times 6 \times 18 + 3 \times 9 \times 27 + \dots + 10 \times 30 \times 90}$$
 is?

17.) The final score in a St. David's vs. King Edward's is m goals for St. David's and n goals for KES.

The number of possible half time scores is?

18.) P is the remainder when each of the numbers 478, 392 and 263 are divided by N, where N is an integer greater than 1. The value of N-P is?

19.) Two circles of equal radii touch as shown. O is the centre of the first circle. OP is tangent to the second circle at P and has length 15. The radius of each circle is?



20.) ABCD is a square. E and F are midpoints of AD and AB as shown. If the area of triangle DEG is 1, then the area of ABCD is?

